Hoang Hieu Long

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Profile

I am an electrical engineer with five years of experience across renewable energy consultancy and power-system research, specialising in HV cable systems, substation concepts and power-system studies, with a working knowledge of energy markets and policy. At OWC – an Aqualis Company, I have delivered electrical design, due diligence and route-to-market studies for 20+ offshore and onshore wind, solar PV, BESS and interconnector projects from 50 MW to 2 GW, led three electrical work packages, one rated 9/10 by the client, and developed an inter-array cable layout optimiser that shortens cable length by 10–30%. Alongside my MSc at UCL, I continue to advise OWC on MV/HV design and power-system studies.

Education

University College London (UCL), **United Kingdom** 2026–2027
MSc Power Systems Engineering

Hanoi University of Science and Technology (HUST), **Vietnam** 2020–2025
BSc Electrical Engineering; top 10% of the cohort; graduation thesis 10/10, ranked 1st of 200+ students

Shibaura Institute of Technology (SIT), **Japan** Apr–Nov 2023
Research exchange, Electromagnetic Actuator Laboratory

Professional and Research Experience

Industry

Consultant – Electrical Engineer, OWC – an Aqualis Company, remote 2026–Present

- Provide HV and MV analysis and advice on renewable energy projects in owner's engineer, technical due diligence and power-system study roles; maintain the team's analytical tools.

Electrical Engineer, OWC – an Aqualis Company, Vietnam 2024–2026

- **Scope:** designed and supported 20+ utility-scale projects (50 MW to 2 GW) in offshore wind, solar PV, BESS and HVAC interconnectors across the UK, Europe, MENA, APAC and the US, from feasibility to FEED.
- **Cable systems and substations:** delivered electrical design for 33 kV PV and BESS systems, 66–132 kV inter-array systems and export cable systems up to 275 kV, plus concept design of 230–500 kV offshore and onshore substations.
- **Power-system studies:** modelled reactive power compensation and losses for HVAC submarine cables over 100 km in DIgSILENT PowerFactory, for interconnector and offshore wind projects.
- **Techno-economics:** delivered CAPEX estimates, NPV-based design optioneering, technical due diligence and route-to-market risk assessments.
- **Leadership:** led 3 electrical work packages, coordinating 5+ multidisciplinary team members.
- **Engineering tools:** built 3 internal tools in Python, VBA and QGIS (IAC layout optimiser, microgrid failure assessment, Power-to-X emissions calculator), presented at WindEurope and used on 5+ live projects; quality-checked 5 in-house design tools used on 10+ projects a year.

Research

MSc Research, University College London 2026–Present

- **Thesis:** Python and web-GIS framework for early-stage grid-connection cost estimation in Great Britain and Australia, linking route screening, HVAC electrical design and lifecycle costing with P10/P50/P90 uncertainty, validated against documented projects.
- **Book chapter** (first author, accepted), ASEAN Centre for Energy: offshore wind risk framework and financial model testing how grid delay, curtailment, offtake price and financing affect NPV and required tariff in Viet Nam and the Philippines.

Hydrogen Team Leader, 100% Renewable Energy Laboratory, HUST 2022–2025
Supervisor: Associate Professor Nguyen Duc Tuyen

- Led a five-member team that published 2 journal articles and 3 conference papers on microgrid optimisation, energy management and PV–electrolyser direct coupling; won USD 5,000 of Green In funding for a green city model with hydrogen storage.
- **Energy management:** developed MATLAB models of an AC microgrid with battery–hydrogen storage and grid-forming control, including a state-space model for model predictive control (MPC) benchmarked against fuzzy-logic and state-machine strategies.
- **Hydrogen systems:** reviewed sizing and operating strategies for direct-coupled PV–electrolyser systems; developed a Power-to-Hydrogen life-cycle assessment method.

Selected Projects

1.32 GW offshore wind, Poland – *lead engineer, inter-array cable package* FEED
Led the team delivering the inter-array cable (IAC) package and liaised with subcontractors and the client; set turbines per IAC string from electrical losses and the cable schedule, and compared IAC designs with and without an offshore substation on NPV. Client feedback 9/10.

500–1,000 MW offshore wind, Morocco – *lead engineer, electrical work package* Feasibility
Led the electrical team and coordinated subcontractor and client inputs; designed IAC systems for four turbine-model and layout options at 66 kV and 132 kV, and compared them on electrical losses, cable schedule and NPV.

Up to 1,200 MW offshore wind, South Korea – *export system optioneering* Pre-FEED
Compared three export concepts for about 150 km of export cable on technical risk and cost-benefit: HVAC export via several offshore substations, HVAC export via a single offshore substation, and HVDC export.

400 MW Indonesia–Singapore HVAC interconnector – *main electrical designer* Interconnector
Designed the HVAC submarine cable system for the 100+ km route and modelled its reactive power compensation and transmission losses in DIgSILENT PowerFactory.

IAC layout optimiser – *developer and project manager, Python* OWC in-house tool
Took the tool from specification to release: a biased genetic algorithm (matrix operations, multiprocessing) that minimises IAC length under design constraints, producing a layout in 2–5 hours instead of days, with 10–30% shorter cable on past projects.

2 GW floating offshore wind, the Philippines – *route-to-market assessment* Market study
Assessed the supply–demand balance and transmission curtailment risk to test whether the project could feasibly connect to the grid, then turned the findings into an investment-risk table and alternative route-to-market options.

50 MW onshore wind farm, Vietnam – *lead electrical reviewer* Technical due diligence
Reviewed the grid-connection design and identified network overloads under normal and N-1 contingency conditions, with recommendations ahead of the investment decision.

Hydrogen strategy benchmarking for Vietnam – *author, 100% Renewable Energy Laboratory* Policy research
Authored 5 policy and market briefs that benchmarked national hydrogen strategies (US, Germany, Japan, China) on long-term targets, demand creation, investment de-risking and R&D support, and set out policy options for Vietnam, including emissions trading, tax incentives and the cost case for hydrogen co-firing in gas plants.

Publications

1. **Long HH**, Huong HL, Giang NH, and Tuyen ND. "From Offshore Wind Pipeline to Bankable Power: Managing Grid, Curtailment and Offtake Risk in ASEAN." Book chapter, *ASEAN Centre for Energy*, 2026. (Accepted)
2. Long PV, **Long HH**, and Tuyen ND. "A Comprehensive Review of Direct Coupled Photovoltaic–Electrolyser System: Sizing Techniques, Operating Strategies, Research Progress, Current Challenges, and Future Recommendations." *International Journal of Hydrogen Energy* (Q1), 48, 25231–25249, 2023.
3. Long PV, **Long HH**, Kien DC, Hirotaka T, and Tuyen ND. "An Improved State Machine-Based Energy Management Strategy for Renewable Energy Microgrid with Hydrogen Storage System." *Energy Reports* (Q1), 9 (Suppl. 1), 194–201, 2023.
4. **Long HH**, Long PV, and Nguyen D. "Power to Hydrogen Life Cycle Assessment: Method and a Case Study for the United States." *WindEurope Annual Event*, Copenhagen, 2025.
5. **Long HH**, Anh NH, Hirotaka T, Goro F, and Tuyen ND. "A Small Scale Microgrid Experimental Model Integrated Battery–Hydrogen Storage System." *IEEJ* Conference, Japan, 2024.
6. Kien DC, Anh NH, **Long HH**, Long PV, Hirotaka T, and Tuyen ND. "Grid Forming-Based Hydrogen System for High-Penetration Renewable Energy Microgrid." *Int. Conf. on Power and Energy Systems Engineering (CPESE)*, Japan, 2024.

Skills and Tools

Power-system analysis:	Reactive power and loss studies, microgrid control – *PowerFactory, MATLAB/Simulink*
Electrical design:	Cable systems, substation concepts, BESS sizing – *QGIS, AutoCAD, Visio*
Techno-economics:	CAPEX, NPV option studies, due diligence – *Excel/VBA cost models*
Programming:	Optimisation, automation and data analysis – *Python, VBA, MATLAB; VS Code, Git*
Reporting:	Technical reports and presentations – *Microsoft 365 (Word, Excel, PowerPoint), LaTeX*
Languages:	English (fluent), Vietnamese (native)

Honours and Activities

OWC Scholarship, top 5 of 200 students in the School 2024
Third Prize, Student Forum, top 30 of 1,000+ student research papers in Vietnam 2022
Delegate, Youth Engagement in Just Energy Transition, UNDP Vietnam 2023
Secretary, HUST Youth Union 2020–2024
Third Prize, Vietnam Mathematical Olympiad, top 45 of 2,000+ in the national round 2020